

THE STRUCTURE OF MATHEMATICAL REALITY

Phase IV — Chapter 12

Information Theory and the Cost of Description

Parallel Reading

Self-Study Curriculum

Parallel Reading: David MacKay, *Information Theory, Inference and Learning Algorithms* (Cambridge, 2003), Chapters 1-4 (approximately 80 pages). This book is freely available online at the author's website. It is rigorous but readable and full of examples.

Also read: Claude Shannon, "A Mathematical Theory of Communication" (1948): read only the introduction and Part I (sections 1-6, about 20 pages). This is one of the most important papers of the 20th century, and it is clearer than most textbooks.